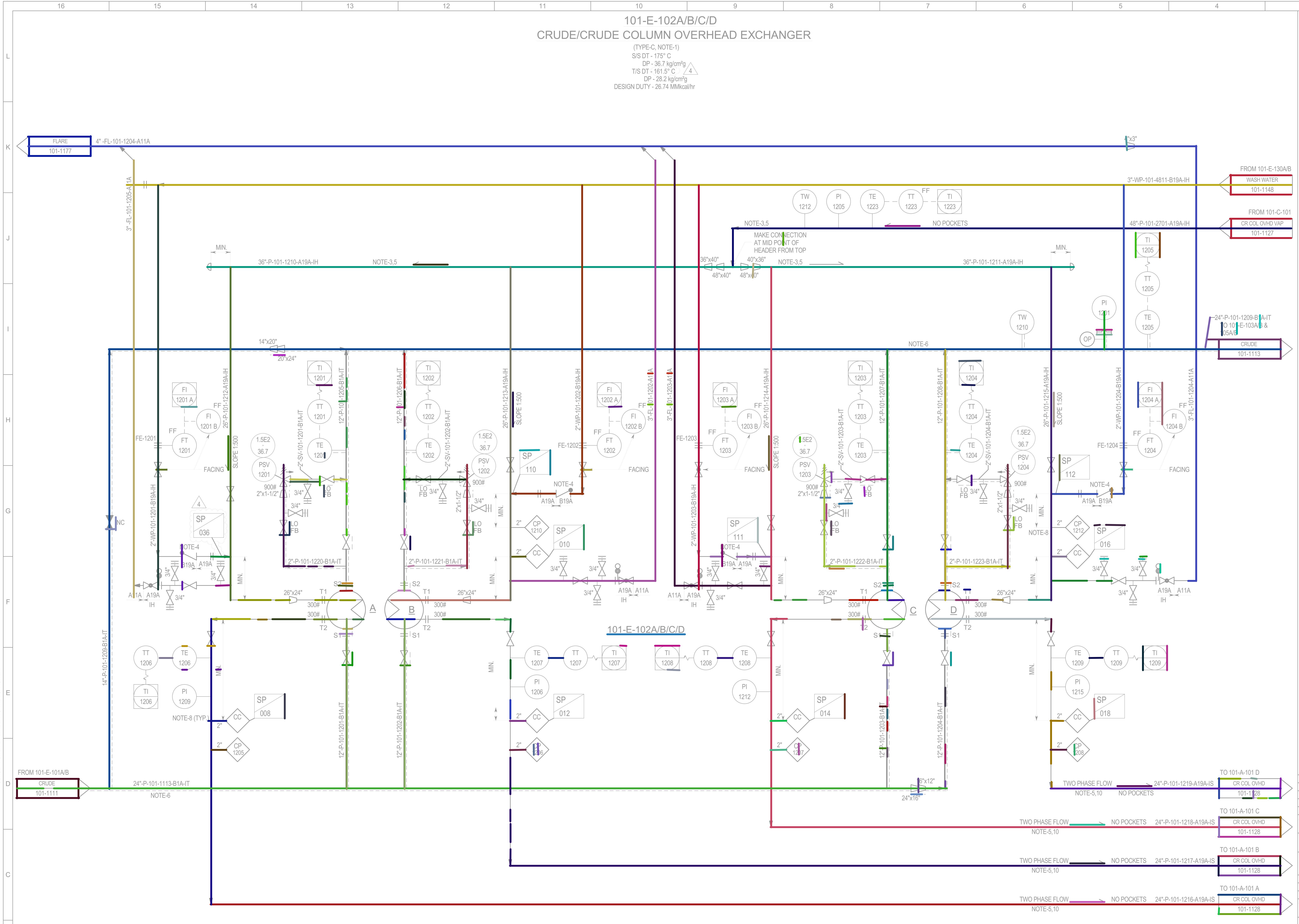
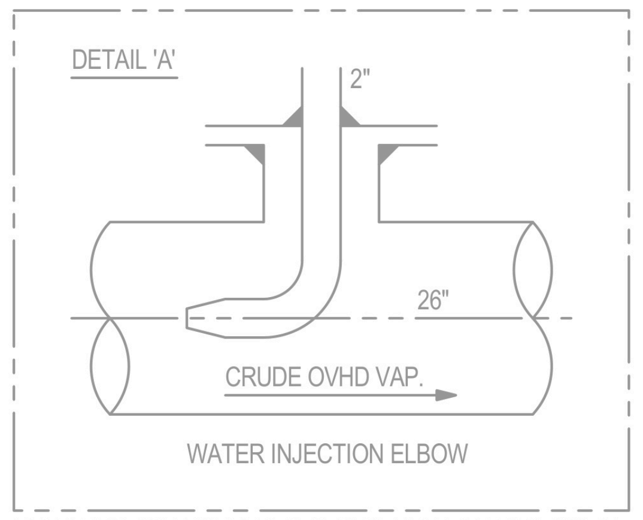
NOTICE: THIS DRAWING AND THE DESIGN IT COVERS ARE THE PROPERTY OF ENGINEERS INDIA LIMITED. THEY ARE MERELY LOANED AND ON THE BORROWER'S EXPRESS AGREEMENT THAT THEY WILL NOT BE REPRODUCED, COPIED, LOANED, EXHIBITED, NOR USED EXCEPT IN THE LIMITED WAY AND PRIVATE USE PERMITTED BY ANY WRITTEN CONSENT GIVEN BY LENDER TO THE BORROWER.


101-E-102A/B/C/D
CRUDE/CRUDE COLUMN OVERHEAD EXCHANGER
(TYPE-C, NOTE-1)
S/S DT - 175° C
DP - 36.7 kg/cm²g
T/S DT - 161.5° C
4
DP - 28.2 kg/cm²g
DESIGN DUTY - 26.74 MMkcal/hr
FLARE
101-1177
4"-FL-101-1204-A11A
FROM 101-E-130A/B
WASH WATER
101-1148
3"-WP-101-4811-B19A-IH
FROM 101-C-101
CR COL OVHD VAP
101-1127
48"-P-101-2701-A19A-IH
NOTE-3,5
NO POCKETS
MAKE CONNECTION AT MID POINT OF HEADER FROM TOP
36"x40"
40"x36"
48"x40"
36"-P-101-1210-A19A-IH
36"-P-101-1211-A19A-IH
MIN.
24"-P-101-1209-B1A-IT
TO 101-E-103A/B & 105A/B
CRUDE
101-1113
NOTE-6
14"x20"
20"x24"
NC
FE-1201
FE-1202
FE-1203
FE-1204
FACING
SLOPE 1:500
NOTE-4
A19A
B19A
A11A
IH
1.5E2
36.7
900#
2"x1-1/2"
LO
FB
3/4"
SP
036
110
010
111
112
016
008
012
014
018
CP 1205
CP 1210
CP 1207
CP 1212
CP 1208
CC
2"
26"x24"
A
B
C
D
S1
S2
T1
T2
300#
101-E-102A/B/C/D
FI 1201 A
FI 1201 B
FT 1201
TI 1201
TT 1201
TE 1201
PSV 1201
PSV 1202
PSV 1203
PSV 1204
TW 1212
PI 1205
TE 1223
TT 1223
TI 1223
TW 1210
PI 1201
TI 1205
TT 1205
TE 1205
TT 1206
TE 1206
TI 1206
PI 1209
TE 1207
TT 1207
TI 1207
PI 1206
TI 1208
TT 1208
TE 1208
PI 1212
TE 1209
TT 1209
TI 1209
PI 1215
NOTE-8 (TYP.)
FROM 101-E-101A/B
CRUDE
101-1111
24"-P-101-1113-B1A-IT
6"x12"
24"x16"
TO 101-A-101 D
TWO PHASE FLOW
NOTE-5,10
NO POCKETS
24"-P-101-1219-A19A-IS
CR COL OVHD
101-1128
TO 101-A-101 C
24"-P-101-1218-A19A-IS
TO 101-A-101 B
24"-P-101-1217-A19A-IS
TO 101-A-101 A
24"-P-101-1216-A19A-IS

NOTES: -

1. FOR LEGEND SYMBOL AND GENERAL DETAILS, REFER P&ID NO. B068-79-41-00-1191/1192. (TPL DWG. NO. B224-CDU-TPL-000-1191/1192-SHT01/1192-SHT02).
   - FOR EXCHANGER DRAIN DETAILS REFER P&ID NO. B224-CDU-TPL-101-1246-SHT01 & B224-CDU-TPL-101-1246-SHT02.
   - FOR AIR COOLER DRAIN CONNECTION DETAILS REFER P&ID NO. B224-CDU-TPL-101-1247.
   - FOR PUMP DRAIN/VENT REFER P&ID NO.B224-CDU-TPL-101-1245-SHT01 & B224-CDU-TPL-101-1245-SHT02.
   - FOR FILTER DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1248.
   - FOR PUMP SEAL PLAN DETAIL REFER P&ID NO. B224-CDU-TPL-101-1184-SHT01, B224-CDU-TPL-101-1184-SHT02 & B224-CDU-TPL-101-1184-SHT03.
   - FOR CONTROL VALVE DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1243-SHT01 & B224-CDU-TPL-101-1243-SHT02.
   - FOR LEVEL INSTRUMENT DRAIN VENT DETAILS REFER P&ID NO. B224-CDU-TPL-101-1244-SHT01 & B224-CDU-TPL-101-1244-SHT02.
   - ALL INSTRUMENT TAG NUMBERS TO HAVE THE UNIT DESIGNATION PREFIX 101.
2. ALL DCS OPEN AND CLOSED LOOP INSTRUMENTS SHOWN IN P&ID WITH `FF' ARE FOUNDATION FIELD BUS DEVICES. FOR INSTRUMENTS FOR WHICH FIELD BUS DEVICES ARE NOT SHOWN/AVAILABLE, SMART TRANSMITTERS/SMART POSITIONERS SHALL BE USED.
3. SYMMETRICAL PIPING.
4. WASH WATER TO BE INJECTED WITH REVERSE FLOW, CONE SPRAY INJECTION NOZZLE, AS PER DETAIL SHOWN BELOW.


DETAIL 'A'
2"
26"
CRUDE OVHD VAP.
WATER INJECTION ELBOW

THE INJECTION NOZZLE TO BE INSERTED IN HORIZONTAL PORTION FOR PROPER MIXING

5. MAXIMUM ALLOWABLE PRESSURE DROP IN THE LINES IN THE CRUDE COLUMN OVHD CIRCUIT (I.E. FROM 101-C-101 TO 101-V-101 VIA 101-E-102A/B/C/D AND 101-A-101) IS 0.1 KG/CM2. ALL LINE SIZES IN THE CIRCUIT ARE CONFIRMED BASED ON FINAL ISOMETRICS.
6. CRUDE INLET AND OUTLET LINES TO BE ORIENTED SUCH THAT THE INLET AND OUTLET ARE DIAGONALLY OPPOSITE TO EACH OTHER AS SHOWN ON P&ID.
7. CORROSION MONITORING SYSTEM (CMS) SHALL BE DONE WITH CORROSION PROBE & CORROSION COUPON.
8. A MIN. OF 1 TO 1.5 M DISTANCE SHOULD BE KEPT BETWEEN CP AND CC. FOR MAINTENANCE PURPOSE.
9. CHEMICAL INJECTION SHALL BE DOSED THROUGH INJECTION TUBE WITH T-TYPE ACCESS FITTINGS.
10. LINE TO BE PROPERLY SUPPORTED.

संख्या No.	संशोधन REVISIONS	तिथि DATE	ड्रॉन DRN	द्वारा BY	जॉच CKD	अनुमोदित APPD
4	ISSUED FOR CONSTRUCTION	10.03.22	SC	MW	VD	PV
3	ISSUED FOR CONSTRUCTION	30.09.21	SC	MW	VD	PV
2	ISSUED FOR ENGG/ DESIGN WITH HAZOP RECOMMENDATION	02.09.20	RMB	KG	RS	SG
1	ISSUED FOR ENGINEERING/DESIGN	11.08.20	JO	KG	RS	SG
0	ISSUED FOR REVIEW	13.03.20	LC	KG	RS	SG

DETAILED ENGINEERING CONTRACTOR: FLUOR®
FLUOR DANIEL INTERNATIONAL
(MALAYSIA) SDN BHD

LSTK CONTRACTOR: TATA
TATA PROJECTS LIMITED
MUMBAI

संख्या No.	तिथि DATE	संशोधन REVISIONS	ड्रॉन DRN	द्वारा BY	जॉच CKD	अनुमोदित APPD
1	14.12.18	REVISED AS MARKED	PP	UA/AT	GN	AM
0	15.11.18	ISSUED FOR PACKAGE	S.K.L	UA/AT	GN	AM

PMC :
इंजिनियर्स इंडिया लिमिटेड
(भारत सरकार का उपक्रम)
ENGINEERS INDIA LIMITED
(A Govt. of India Undertaking)

OWNER
एच पी सी एल राजस्थान रिफाइनरी लिमिटेड
HPCL RAJASTHAN REFINERY LTD
RAJASTHAN REFINERY PROJECT

पाइपिंग एण्ड इंस्ट्रूमैन्टेशन डायाग्राम
PIPING AND INSTRUMENTATION DIAGRAM
CRUDE/VACUUM DISTILLATION UNIT
CRUDE PREHEAT TRAIN-1B

अनुमाप SCALE	कार्य संख्या JOB NO.	विभाग DEPT.	अनुभाग SECTN.	इकाई UNIT	आरेख संख्या DWG. No.	संशो REV.
∽	B068	79	41	101	1112	1

TPL DRAWING NO. टी पी एल आरेख संख्या: B224-CDU-TPL-101-1112 — संशो REV 4

DOCUMENT REVIEW DETAILS

☑ 1 No Comments. **Doc. B224-CDU-TPL-101-1112 Rev. 4**
☐ 2 Proceed with manufacture / fabrication as per the commented document. Revised document required.
☐ 3 Document does not conform to basic requirements as marked. Resubmit for review.
☐ R Retained for Records. ☐ V Void.

Signature with Date: 24-AUG-2022
Name of Reviewer: UMER AFZAL (C889)
Page No. 1 of 2

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